Math 257 Definitions and Theorems

Week 7: Modules 22-25

Fall 2023

22 Coordinates

Theorem 22.1. *Let $(\mathbf{v}_1, \ldots, \mathbf{v}_p)$ be a basis of V. Then every vector $\mathbf{w}$ in V can be expressed **uniquely** as*

$$\mathbf{w} = c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p.$$

Definition 22.1. Let $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p)$ be an (ordered) basis of V, and let $\mathbf{w} \in V$. The **coordinate vector $\mathbf{w}_\mathcal{B}$** of $\mathbf{w}$ with respect to the basis $\mathcal{B}$ is

$$\mathbf{w}_\mathcal{B} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_p \end{bmatrix}, \quad \text{if } \mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p.$$

Definition 22.2. In $\mathbb{R}^n$ let $\mathbf{e}_i$ denote the vector with a 1 in the i-th coordinate and 0's elsewhere. The **standard basis of** $\mathbb{R}^n$ is the ordered basis $\mathcal{E}_n := (\mathbf{e}_1, \ldots, \mathbf{e}_n)$.

Definition 22.3. Let $\mathcal{B}$ and $\mathcal{C}$ be two bases of $\mathbb{R}^n$. The **change of basis matrix** $I_{\mathcal{C},\mathcal{B}}$ is the matrix such that for all $\mathbf{v} \in \mathbb{R}^n$

$$I_{\mathcal{C},\mathcal{B}}\mathbf{v}_\mathcal{B} = \mathbf{v}_\mathcal{C}$$

Theorem 22.2. *Let $\mathcal{B} = (\mathbf{b}_1, \ldots, \mathbf{b_n})$ be a basis of $\mathbb{R}^n$. Then*

$$I_{\mathcal{E}_n,\mathcal{B}} = \begin{bmatrix} \mathbf{b}_1 & \ldots & \mathbf{b}_n \end{bmatrix}$$

That is, for all $\mathbf{v} \in \mathbb{R}^n$,

$$\mathbf{v} = \begin{bmatrix} \mathbf{b}_1 & \ldots & \mathbf{b}_n \end{bmatrix} \mathbf{v}_\mathcal{B}.$$

Remark. Let $\mathcal{B}$ and $\mathcal{C}$ be bases in $\mathbb{R}^n$.

- To compute $I_{\mathcal{B},\mathcal{E}_n}$, notice that

$$\mathbf{v} = I_{\mathcal{E}_n,\mathcal{B}}\mathbf{v}_\mathcal{B} \rightsquigarrow I^{-1}_{\mathcal{E}_n,\mathcal{B}}\mathbf{v} = \mathbf{v}_\mathcal{B} \rightsquigarrow I_{\mathcal{B},\mathcal{E}_n} = I^{-1}_{\mathcal{E}_n,\mathcal{B}}$$

- To compute $I_{\mathcal{B},\mathcal{C}}$, we can either use

$$I_{\mathcal{B},\mathcal{E}_n} I_{\mathcal{E}_n,\mathcal{C}} \mathbf{v}_{\mathcal{C}} = I_{\mathcal{B},\mathcal{E}_n} \mathbf{v} = \mathbf{v}_{\mathcal{B}} \rightsquigarrow I_{\mathcal{B},\mathcal{C}} = I_{\mathcal{B},\mathcal{E}_n} I_{\mathcal{E}_n,\mathcal{C}}$$

  or use

$$I_{\mathcal{C},\mathcal{B}} = \begin{bmatrix} (\mathbf{b}_1)_{\mathcal{C}} & \dots & (\mathbf{b}_n)_{\mathcal{C}} \end{bmatrix}$$

23 Orthogonal and Orthonormal Bases

Theorem 23.1. *Let $\mathbf{v}_1, \dots, \mathbf{v}_m \in \mathbb{R}^n$ be non-zero and pairwise orthogonal. Then $\mathbf{v}_1, \dots, \mathbf{v}_m$ are linearly independent.*

Remark. The theorem implies that a set of n orthonormal vectors in $\mathbb{R}^n$ is a basis of $\mathbb{R}^n$.

Definition 23.1. An **orthogonal basis** (an **orthonormal basis**) is an orthogonal set of vectors (an orthonormal set of vectors) that forms a basis.

Remark. When $\mathcal{B}$ is orthonormal, then $\mathbf{b}_i \cdot \mathbf{b}_i = 1$ for $i = 1, \dots, n$.

Theorem 23.2. *Let $\mathcal{B} := (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ be an orthogonal basis of $\mathbb{R}^n$, and let $\mathbf{v} \in \mathbb{R}^n$. Then*

$$\mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{b}_1}{\mathbf{b}_1 \cdot \mathbf{b}_1} \mathbf{b}_1 + \dots + \frac{\mathbf{v} \cdot \mathbf{b}_n}{\mathbf{b}_n \cdot \mathbf{b}_n} \mathbf{b}_n.$$

Theorem 23.3. *Let $\mathcal{U} = (\mathbf{u}_1, \dots, \mathbf{u_n})$ be an orthonormal basis of $\mathbb{R}^n$. Then*

$$I_{\mathcal{U},\mathcal{E}_n} = \begin{bmatrix} \mathbf{u}_1 & \dots & \mathbf{u}_n \end{bmatrix}^T.$$

Definition 23.2. An $n \times n$-matrix Q is **orthogonal** if $Q^{-1} = Q^T$.

24 Linear Transformations

Definition 24.1. Let V and W be vector spaces. A map $T : V \to W$ is a **linear transformation** if

$$T(a\mathbf{v} + b\mathbf{w}) = aT(\mathbf{v}) + bT(\mathbf{w})$$

for all $\mathbf{v}, \mathbf{w} \in V$ and all $a, b \in \mathbb{R}$.

Remark. $T(\mathbf{0}_V) = T(0 \cdot \mathbf{0}_V) = 0 \cdot T(\mathbf{0}_V) = \mathbf{0}_W \rightsquigarrow T(\mathbf{0}_V) = \mathbf{0}_W$

Theorem 24.1. *Let V, W be two vector spaces, let $T : V \to W$ be a linear transformation and let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V. Then T is completely determined by the values $T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)$.*

Theorem 24.2. *Let $T \colon \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then there is a $m \times n$ matrix A such that*

- $T(\mathbf{v}) = A\mathbf{v}$, *for all* $\mathbf{v} \in \mathbb{R}^n$.
- $A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix}$, *where* $(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$ *is the standard basis of* $\mathbb{R}^n$.

Remark. We call this A the **coordinate matrix of** T with respect to the standard bases - we write $T_{\mathcal{E}_m, \mathcal{E}_n}$.

25 Coordinate Matrices of Linear Transformations

Theorem 25.1. *Let* V, W *be two vector space, let* $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ *be a basis of* V *and* $\mathcal{C} = (\mathbf{c}_1, \dots, \mathbf{c}_m)$ *be a basis of* W*, and let* $T \colon V \to W$ *be a linear transformation. Then there is a* $m \times n$ *matrix* $T_{\mathcal{C},\mathcal{B}}$ *such that*

- $T(\mathbf{v})_{\mathcal{C}} = T_{\mathcal{C},\mathcal{B}} \mathbf{v}_{\mathcal{B}}$, *for all* $\mathbf{v} \in V$.
- $T_{\mathcal{C},\mathcal{B}} = \begin{bmatrix} T(\mathbf{b}_1)_{\mathcal{C}} & T(\mathbf{b}_2)_{\mathcal{C}} & \dots & T(\mathbf{b}_n)_{\mathcal{C}} \end{bmatrix}$.

Theorem 25.2. *Let* $T : \mathbb{R}^m \to \mathbb{R}^n$ *be a linear transformation and* $\mathcal{A}$ *and* $\mathcal{B}$ *be two bases of* $\mathbb{R}^m$ *and* $\mathcal{C}, \mathcal{D}$ *be two bases of* $\mathbb{R}^n$*. Then*

$$T_{\mathcal{C},\mathcal{A}} = I_{\mathcal{C},\mathcal{D}} T_{\mathcal{D},\mathcal{B}} I_{\mathcal{B},\mathcal{A}}.$$